Nonlinear Dynamics & Chaos HW1

Daniel Haim Breger, 316136944

Technion

(Date: May 22, 2026)

1 Phase-space contraction of the damped pendulum

The damped pendulum obeys

$$\ddot{\theta} = -\gamma\dot{\theta} - \frac{g}{L}\sin\theta, \qquad \gamma > 0 \tag{1}$$

1. Rewrite the equation as a first-order system in the phase-space coordinates. Check if the phase space area is conserved by the flow. Does the answer depend on the point you consider?
2. Describe qualitatively, in a short paragraph, how the pendulum moves in time for different initial conditions. Sketch the corresponding phase portrait in the (θ, ν) plane: identify the fixed points, indicate which are stable and which are unstable, and draw a few representative trajectories. How does the addition of dissipation change the phase space portrait?

1.1 First order system

Using the definition $\nu = \dot{\theta}$ we get the system

$$\dot{\theta} = \nu \qquad\qquad \dot{\nu} = -\gamma\nu - \frac{g}{L}\sin\theta \tag{2}$$

To check whether the phase space area is conserved, we can take the divergence of the system

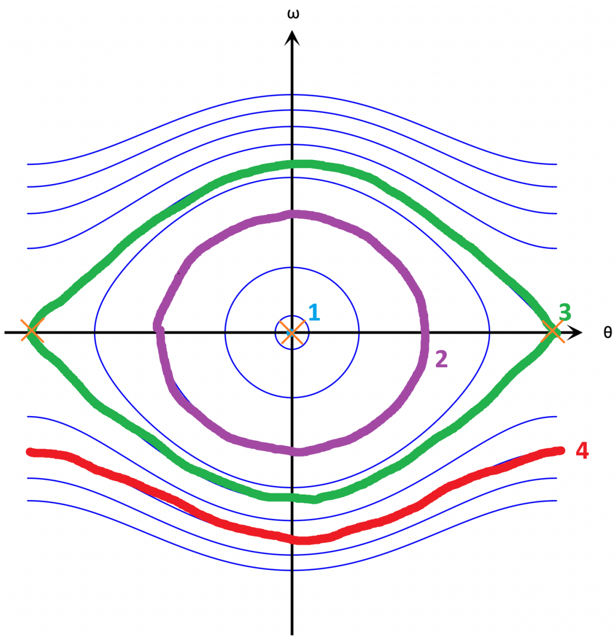
$$\boldsymbol{\nabla}\cdot\mathbf{f} = \boldsymbol{\nabla}\cdot\left(\nu, -\gamma\nu - \frac{g}{L}\sin\theta\right) = 0 - \gamma < 0 \tag{3}$$

So the phase space area shrinks. This does not have any dependence on the phase space point at which we're at (with the trivial exception of (0,0)). This also makes sense, as a damped pendulum will always lose energy and tend to eventually rest.

1.2 Qualitative Explanation

I am assuming most of this subsection deals with an undamped pendulum. I also apologize for the drawings being made in paint, I did not have access to my iPad.

There are 4 sets of starting conditions which can be viewed as different. The trivial is that the pendulum begins at (0,0) in phase space (marked 1 in the sketch), in which case it remains static at rest over time. The next starting conditions is in the area where the pendulum's motion can be well approximated with the small angle approximation (marked 2 in the sketch). There the pendulum behaves as we know pendulums. The third set of starting conditions is such that the phase space line for the movement is still closed by the angles reached by the pendulum are too great for the small angle approximation (marked 3 in the sketch). In this case the pendulum still rocks back and forth, but in a more complex way. The last set of starting conditions is such that the angular velocity of the



pendulum is too great for gravity to overcome it and reverse direction, making the pendulum rotate in one direction (marked 4 in the sketch). The fixed points are marked with an orange X. The fixed point at (0,0) is stable, while the two others are unstable.

Adding dissipation would change the phase space portrait by curving all lines towards the center, disallowing any trajectory which does not eventually tend towards (0,0).

2 Fixed points and stability

1. Consider the system $\dot{x} = \tan x$ restricted to the segment $x \in (-\pi/2, \pi/2)$. Find and classify the fixed points in this segment.
2. Consider the system $\dot{x} = \left(1 - e^{-x^2}\right)(1 - x)$
   (a) Find and classify the fixed points of the system using linear stability analysis. If linear stability analysis fails determine stability using a graphical method instead.
   (b) For an iitial condition very close to the stable fixed point, find how $x(t)$ behaves at long times.
3. Consider the system $\dot{x} = -x^3$
   (a) Find and classify the fixed points of the system
   (b) Solve this equation exactly for a general initial condition and describe how $x(t)$ behaves at long times. Compare with the long-time behavior you found. What is qualitatively differnent and why?

2.1 tanx

The fixed points can be found by solving

$$\tan x = 0 \implies x = k\pi \quad k \in \mathbb{Z} \tag{4}$$

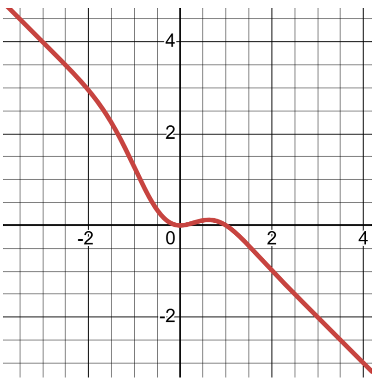
therefore the only fixed point in the segment is (0,0), and it is unstable.

2.2 Gaussian and linear

2.2.1 Fixed Points

To begin the process we find points where we suspect there may be fixed points. From the form of the equation we can immediately tell that $x = 0, 1$ are fixed points which we will investigate. To perform linear stability analysis we expand f in a taylor series around those points. First we derive

$$f'(x) = \left(2xe^{-x^2}\right)(1 - x) - 1 + e^{-x^2} = \begin{cases} 0 & x = 0 \\ \frac{1}{e} - 1 & x = 1 \end{cases} \tag{5}$$



Therefore the point $x = 1$ is stable, and the point $x = 0$ requires a graphical analysis. Looking at the graph of the function we can see that at the point $x = 0$ going to the negative gives positive values for $\dot{x}$, which stabilizes the point, but going to positive values also gives positive $\dot{x}$, which pushes us away from the point, so $x = 0$ is not stable.

2.2.2 Behavior at long times

The stable point as we found is (1,0).

3 Bifurcations

For each of the following systems, identify which types of bifurcation occur at a critical r: saddle node, transcritical, or pitchfork. Determine the critical r values and sketch the bifurcation diagram x^* vs. r. Draw the flow on the phase line before and after the bifurcation.

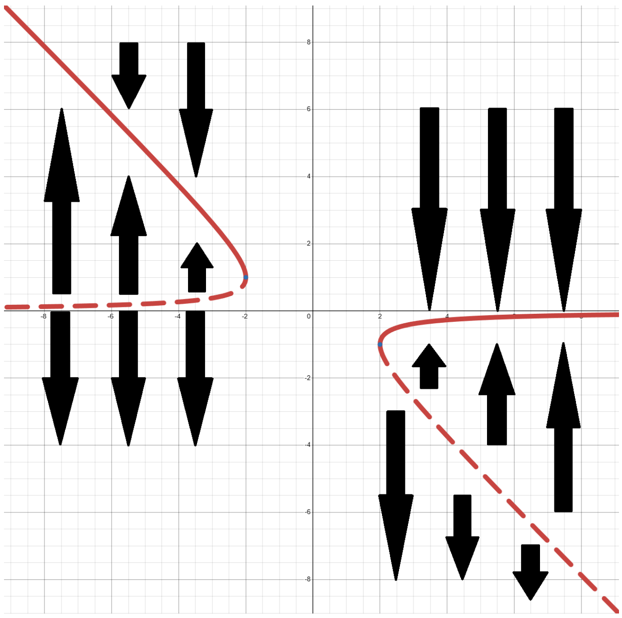
- $\dot{x} = 1 + rx + x^2$
- $\dot{x} = -x + \frac{rx}{1+x^2}$
- $\dot{x} = x\left(r - e^x\right)$
- $\dot{x} = r - \cosh x$

3.1 $\dot{x} = 1 + rx + x^2$

We can find the roots of the right hand side by

$$x^* = \frac{-r \pm \sqrt{r^2 - 4}}{2} = \frac{-r \pm \sqrt{(r+2)(r-2)}}{2} \tag{6}$$

We can see that there are either no critical points (when $-2 < r < 2$), one critical point (at $r = \pm 2$) or two critical points (when $|r| > 2$). Therefore both $r = \pm 2$ are saddle node bifurcations.



3.2 $\dot{x} = -x + \frac{rx}{1+x^2}$

Simplifying the right hand side

$$-x + \frac{rx}{1+x^2} = x\left(\frac{r}{1+x^2} - 1\right) \tag{7}$$

threfore the critical points are at

$$x = 0 \qquad\qquad x = \sqrt{r-1} \tag{8}$$

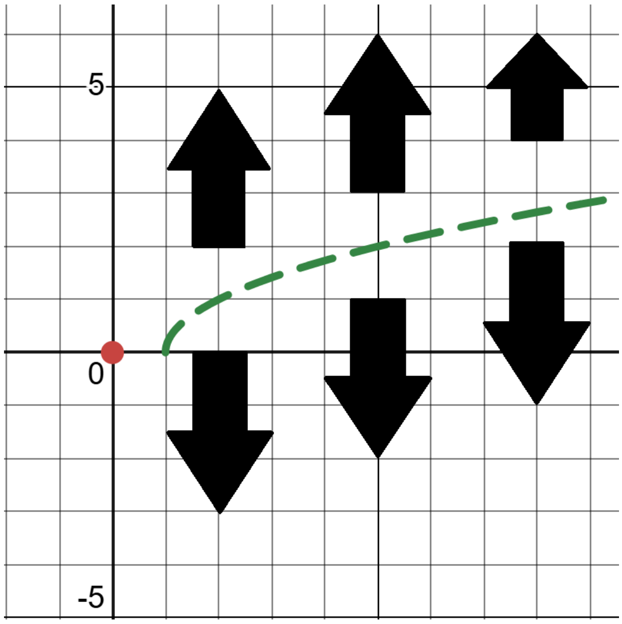
Therefore there is always a critical point at $x = 0$ but also another one which moves with r and appears when $r \geq 1$. Looking at the derivative of the right hand side at $x = 0$

$$f'(0) = -1 + \left.\frac{r\left(1+x^2\right) - 2rx^2}{\left(1+x^2\right)^2}\right|_{x=0} = r - 1 \tag{9}$$

Therefore the critical point at $x = 0$ changes its stability depending on the value of r, therefore this is a transcritical bifurcation. As for the other point

$$f'(\sqrt{r-1}) = -1 + \left.\frac{r\left(1+x^2\right) - 2rx^2}{\left(1+x^2\right)^2}\right|_{x=\sqrt{r-1}} = \frac{2r\,(r-1)}{r^2} = 2\left(1 - \frac{1}{r}\right) \tag{10}$$

Which is always positive for $r \geq 1$, so it is unstable.



3.3 $\dot{x} = x\left(r - e^x\right)$

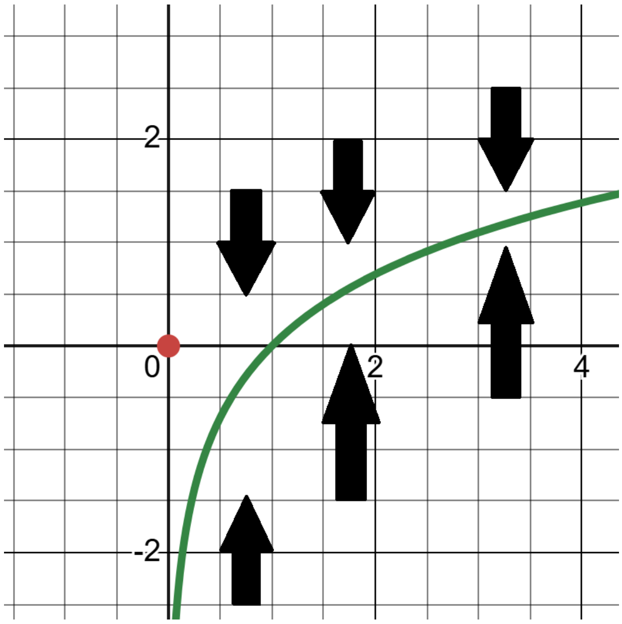
It's easy to see the critical points are $x = 0, \ln r$. The derivative is

$$f'(x) = r - e^x - xe^x = r - e^x\left(1 + x\right) \tag{11}$$

therefore

$$f'(0) = r - 1 \qquad\qquad f'(\ln r) = -r\ln r \tag{12}$$

The critical point at $x = 0$ changes its stability, and the critical point at $x = \ln r$ is always stable. Therefore this is a transcritical bifurcation.

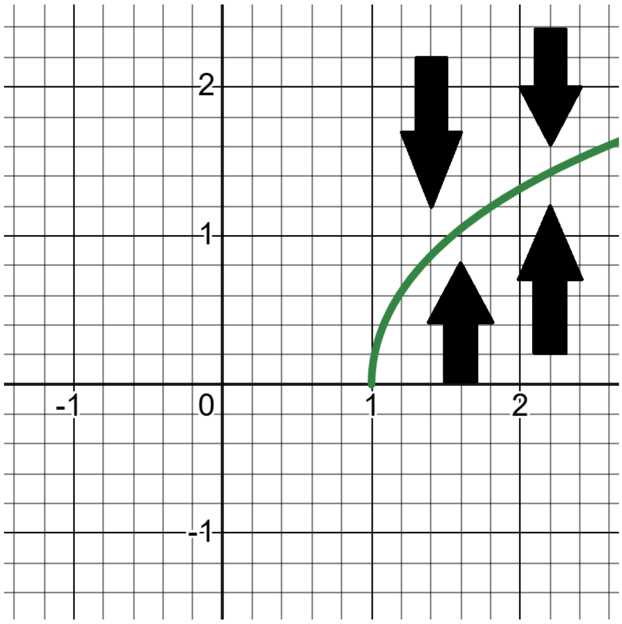


3.4 $\dot{x} = r - \cosh x$

The root of the right hand side is, expectedly, $x = \cosh^{-1}(r)$. The derivative is $f'(x) = -\sinh x$. Therefore the derivative at the critical point is

$$f'(\cosh^{-1}(r)) = -\sinh\left(\cosh^{-1}(r)\right) \leq 0 \quad \text{(thanks desmos)} \tag{13}$$

So the critical point is always stable. This is also a transcritical bifurcation.



4 Numerical exploration: the supercritical pitchfork

Very long description